



Does transparency matter when an AI system meets performance expectations? An experiment with an online dating site

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ABSTRACT

Powered by sophisticated algorithms, artificial intelligence (AI) systems are notorious for being opaque, prompting demands for more algorithmic transparency. However, empirical evidence reveals inconsistent effects of transparency on user trust, suggesting critical boundary conditions. Drawing on the Expectation Confirmation Model and Heuristic-Systematic Model, we propose that the effectiveness of providing algorithmic transparency depends on whether the AI system's performance meets, exceeds, or falls short of users' expectations. We tested this through a 3 (Algorithmic transparency: high vs. low vs. absent) $\times$ 3 (Expectation confirmation: positive vs. simple vs. negative) between-subjects online experiment ($N = 227$), in which participants interacted with an online dating site driven by date-matching algorithms. Results revealed that detailed algorithmic explanations enhanced perceived understanding and trust when system performance either exceeded or fell short of expectations, while concise explanations proved more effective when performance met expectations. These findings advance our theoretical understanding of algorithmic transparency and provide practical implications for designing adaptive explainable AI systems that adjust transparency levels based on performance outcomes.

By leveraging sophisticated algorithms and vast amounts of personal user data, AI-driven platforms have become increasingly “smart” in recommending suitable movies, products, and even social connections. To increase the accuracy of predictions, developers continue to build machine learning models that become increasingly complicated, creating the infamous “black box” problem (Breiman, 2001) wherein users cannot discern how these AI systems generate recommendations. Such limited understandability may potentially lead to algorithmic anxiety (Jhaver et al., 2018) and undermine users' trust (e.g., Toff & Nielsen, 2018). Even when users are familiar with an AI system, they tend to lose trust quickly if an AI system performs worse than expected, contributing to AI aversion (Dietvorst et al., 2015). Thus, although prior work has shown that AI systems can match or surpass human performance in tasks such as medical image classification (Esteva et al., 2017), sales (Luo, Qin, Fang, & Qu, 2021), and matchmaking (Sharabi, 2021), the lack of appropriate understanding and trust has become a major impediment to accepting AI systems' advice (Dietvorst et al., 2015).

These issues have triggered rapid growth in research on algorithmic transparency. Providing users with information on how AI systems generate decisions has been identified as a strategy to help laypersons form a sufficient understanding (Kulesza et al., 2012) and place more trust in the AI systems (e.g., Cramer et al., 2008). Despite these growing efforts, the decision to provide algorithmic transparency has yielded mixed effects on trust outcomes. While some studies demonstrate that algorithmic explanations enhance user trust in AI systems (e.g., Cotter et al., 2017; Molina & Sundar, 2022), others find that transparency has no effect or even undermines trust when users discover unexpected algorithmic behaviors (Jhaver et al., 2018; Kizilcec, 2016). Still other research suggests that users may not require detailed explanations if AI systems work effectively (Bunt et al., 2012). These inconsistent findings suggest that transparency's effectiveness depends on critical but unidentified boundary conditions.

We propose that a key missing factor is users' performance expectations regarding the AI system. Previous research has shown that when

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systems perform better than expected, users tend to trust the systems more (Burgoon et al., 2016), but they quickly lose trust when unexpected errors occur (Dietvorst et al., 2015). Thus, users may only require a more detailed explanation when unexpected outcomes occur to resolve the inconsistency. In fact, users may not require detailed information if AI systems work effectively (Bunt et al., 2012); they have a good understanding with minimal information (Tullio et al., 2007). In this case, providing algorithmic transparency information may create a cognitive burden (Bunt et al., 2012) or even erode users' trust if there is no direct information from the system explaining unexpected performance (Jhaver et al., 2018), especially when information does not help the users understand unexpected results (Kizilcec, 2016).

Drawing on the Expectation Confirmation Model (ECM; Bhattacharjee, 2001), we theorize that expectation disconfirmation moderates the effectiveness of algorithmic transparency on user trust in AI systems. We test our theoretical predictions in the context of AI-mediated online dating, a domain where algorithmic matching procedures play a critical role in helping users navigate vast pools of potential partners (Sharabi, 2021) and shaping AI-mediated communication (Hancock et al., 2020).

Our literature review proceeds in three steps. First, we establish the Expectation Confirmation Model (ECM) as our theoretical foundation, extending it from consumer satisfaction (Bhattacharjee, 2001) to AI trust formation. Second, we examine the literature on algorithmic transparency, identifying 'perceived understanding' as the mechanism linking transparency to trust. Third, we integrate these streams, theorizing that expectation disconfirmation moderates the effectiveness of algorithmic transparency through perceived understanding.

1. Effects of user expectations on trust in AI systems

Trust is an evolving phenomenon, "a dynamic expectation that undergoes predictable changes as a result of an experience in a relationship" (Rempel & Holmes, 1986, p. 20). Researchers have found that the process of establishing trust towards AI systems generally depends on how *reliable* the systems are (Dzindolet et al., 2003; Rempel et al., 1985). Reliability is not purely objective. In fact, it is usually tied to users' expectations. When encountering information about someone or something inconsistent with one's expectancies, individuals are more likely to perceive the target as unreliable and tend to be more uncertain about their trust decisions (Brashers, 2001). Such an effect of perceived reliability on interpersonal trust can also be applied in human-AI interactions. Specifically, previous literature has shown that users' perceptions of whether an automated system performs its functions correctly constitute a crucial determinant of trust (Jian et al., 2000; Muir & Moray, 1996). Furthermore, the more participants believe that they could trust an automated system's judgment, the higher their expectations for the system's performance (Dzindolet et al., 2003). If users detect errors or if the automated aid performs abnormally, users' trust may dwindle since abnormal performances are more likely to be remembered than normal ones (Dzindolet et al., 2003). Therefore, the congruence between expectations and performance is foundational for building users' trust in automation (Lee & Moray, 1992) and a crucial reference point for future use decisions (Anderson & Sullivan, 1993; Bonito et al., 1999; Kaur & Rampersad, 2018).

As most studies of human-AI interaction focus on how the performance of automated systems affects trust formation (e.g., Dzindolet et al., 2003; Jacovi et al., 2021), we lack a firm understanding of how users perceive and form trust when the AI system's performance aligns with or violates users' expectations. In a review of recent empirical research about users' trust in AI, Glikson and Woolley (2020) underscored the importance of examining the consistency, or lack thereof, between users' expectations and the AI system's performance, to understand users' behaviors and determine the best strategy for facilitating trust. In the following section, we draw on the Expectation Confirmation Model to explain how confirmation or disconfirmation of users' expectations may affect their trust in AI systems.

1.1. Expectation Confirmation Model

The Expectation Confirmation Model (ECM) provides a framework for understanding how the interplay between expectations and actual performance shapes user satisfaction and continued use intentions (Bhattacharjee, 2001). Built upon the early research in consumer behavior (Oliver, 1980), which posits that individuals form initial expectations before using a product or service and evaluate perceived performance against these prior expectations, leading to one of three outcomes: confirmation (performance meets expectations), positive disconfirmation (performance exceeds expectations), or negative disconfirmation (performance falls short of expectations). Bhattacharjee (2001) adapted this framework to the context of information systems, proposing that users' satisfaction and continuance intentions are shaped by their prior expectations and confirmation of expectations. The ECM predicts that confirmation and positive disconfirmation lead to satisfaction and continued usage, while negative disconfirmation results in dissatisfaction and discontinuance. This framework has been widely validated across various technology contexts, including e-commerce (Bhattacharjee & Premkumar, 2004), mobile applications (Oghuma et al., 2016), and smart devices (Park, 2020).

1.2. Extending ECM to AI trust formation

Recent studies have begun applying ECM to AI and conversational agent contexts, yielding important insights into how expectation disconfirmation influences user responses. For instance, Burgoon et al. (2016) found that when an embodied agent displayed unexpected but favorable characteristics (i.e., positive disconfirmation), users reported feeling more connected to and understood by the agent than they did when their expectations were negatively violated. Similarly, Rheu et al. (2024) examined chatbots for social support and revealed that positive disconfirmation significantly improved evaluations of the chatbot and emotional validation. Park (2020) further demonstrated that positive confirmation was a key influential factor in users' continuous use of smart wearable devices. In contrast, studies have shown that negative disconfirmation of a chatbot's performance negatively affects user experience (Lee, 2010; Nazaretsky et al., 2021; Wang et al., 2020).

While these studies provide empirical evidence suggesting the important role of expectation disconfirmation in AI contexts, there remains a lack of theory-driven analysis examining how expectation confirmation and disconfirmation specifically influence trust in AI systems (Glikson & Woolley, 2020). When an AI system's performance exceeds expectations (positive disconfirmation), users should experience not only greater satisfaction but also enhanced cognitive trust. Cognitive trust refers to users' beliefs about a system's reliability, competence, and dependability (Mayer et al., 1995). When a system demonstrates capability beyond what was anticipated, it signals greater reliability and competence, thereby strengthening users' confidence in the system. Conversely, when performance falls short (negative disconfirmation), both trust and behavioral intentions should be undermined, as the system fails to meet the baseline standards expected by users.

1.3. Application to AI-mediated dating

In the context of online dating platforms, users approach the experience with preconceived notions about the quantity and quality of potential matches they might encounter. These expectations are shaped by various factors, including past experiences, marketing claims of dating sites, and social comparisons (Sharabi & Caughlin, 2017). Users may anticipate receiving a certain number of "optimal matches" or recommendations (Tong et al., 2016, p. 656), forming relational schemas for online dating interactions. These schemas include expectations about the number of potential partners they might interact with, influencing their strategies for relationship initiation and development (LeFebvre,

2018).

When the actual results deviate significantly from these expectations, it can lead to expectation disconfirmation in the context of online relationship formation (Ramirez et al., 2015). According to ECM, positive disconfirmation (i.e., when the AI dating system provides better matches or more Top Picks than expected) should lead to greater trust in the system's competence and reliability (cognitive trust), as well as stronger intentions to continue using the system. The system has demonstrated capability beyond the baseline expectation, which enhances users' confidence in its algorithmic matching abilities. Conversely, negative disconfirmation (i.e., when matches fall short of expectations) should undermine both trust and usage intentions. This erosion of trust is particularly problematic in the domain of online dating, where users rely heavily on algorithmic recommendations to navigate a vast pool of potential partners (Sharabi, 2021). Therefore, extending ECM to the context of AI-mediated dating systems, we propose the following hypothesis.

H1. When an AI system positively disconfirms users' expectations, users will have (a) greater cognitive trust, and (b) stronger behavioral intentions in using the AI system than a system that simply confirms or negatively disconfirms their expectations of the system's performance.

H1 predicts that expectation disconfirmation affects trust and intentions. However, when systems violate expectations, users face uncertainty about why performance differed from what they anticipated. This raises a critical question: What kind of information would help users process expectation disconfirmation and calibrate trust appropriately? We propose that algorithmic transparency plays this role by providing insight into system operations.

1.4. Effects of transparency on user trust in AI systems

Emerging AI research has empirically demonstrated the value of explaining AI systems to users in combating the "black box" issue of AI (e.g., Molina & Sundar, 2022). A growing number of studies have been devoted to investigating *what* constitutes a good explanation for trust-building (e.g., Anik & Bunt, 2021; Rader et al., 2018). However, how transparency affects user trust may largely depend on how users process such information. According to the heuristic-systematic model (HSM; Chaiken, 1980), individuals apply either systematic or heuristic strategies to process information, or sometimes both, in order to arrive at a judgment. Systematic processing occurs when individuals engage in considerable effort in searching for information and scrutinizing arguments to ensure high quality of the information in decision-making, while heuristic processing occurs when individuals rely on "simple decision rules" to help them make decisions quickly (Gigerenzer & Gaissmaier, 2011).

When interacting with AI systems, users often pay attention to "cues" on the interface that can help them evaluate the extent to which they can trust the system. Thus, when transparency information is available, users may value a system more if it discloses in-depth information compared to a simple, short message, as they apply mental shortcuts to judge a message's credibility based on superficial cues (Sundar, 2008). This is especially the case when interacting with complex AI systems. In line with the "length equals strength" heuristic in HSM (Chaiken, 1980; Chen & Chaiken, 1999), when more information is disclosed to users, users will be inclined to apply the rule of thumb that a lengthy disclosure is a strong disclosure, without reading the information closely (Sundar et al., 2020). Existing research has demonstrated the effectiveness of the mere presence of algorithm transparency, and users trust a system more if it provides detailed information (see Sundar et al., 2020).

In online dating, where algorithms play a crucial role in matching potential partners, users often desire insight into how these systems make decisions (Tong et al., 2016). Disclosing information about an AI system's operations to users can also help users derive meaning from AI systems and amend their trust in the system when unexpected

performance is detected (Brashers, 2001). Therefore, perceived understanding, or the extent to which users perceive themselves as having gained an understanding of AI systems, has been shown to increase users' trust and satisfaction (see Eslami et al., 2016). Considering the empirical findings, we propose the following hypotheses.

H2. Providing more (vs. less) transparency will enhance users' perceived understanding of the AI system.

H3. Perceived understanding of the AI system will mediate the effects of transparency on a) cognitive trust, and (b) behavioral intentions in using the AI system in the future.

Thus far, we have established two independent pathways: expectation disconfirmation affects trust and intentions (H1), while algorithmic transparency enhances understanding and subsequently trust (H2-H3). However, these pathways may not operate independently. We propose that the effectiveness of algorithmic transparency depends critically on whether users' expectations are met or violated. This interaction is theoretically grounded in dual-process models of social cognition and trust formation.

1.5. The interplay between performance expectation and transparency

The relationship between expectation disconfirmation and transparency in AI systems is theoretically grounded in both cognitive processing and trust formation frameworks. Understanding a system's decision-making process requires cognitive effort, which becomes particularly crucial when system performance deviates from expectations (Lee & Moray, 1992). This aligns with the sufficiency principle in the Heuristic-Systematic Model (HSM), which posits that individuals engage in systematic processing when they perceive a need for thorough evaluation (Chaiken, 1980; Chen & Chaiken, 1999). When users encounter expectation disconfirmation, they typically transition from heuristic to systematic processing modes, triggering increased attention and information-seeking behaviors (Dzindolet et al., 2003).

Research has demonstrated that expectation disconfirmation catalyzes heightened cognitive engagement, prompting users to seek explanations for unexpected outcomes (Roese & Sherman, 2007). This increased cognitive involvement suggests that algorithmic transparency becomes more salient and potentially more impactful during instances of expectation disconfirmation. This study advances the conceptualization that the disconfirmation of expectations may serve as the condition that catalyzes the need for more information and systematic information processing: when users perceive discrepancies between their expectations and a system's performance, they are likely to switch their mode of processing information from the heuristic route to the systematic route (Dzindolet et al., 2003). For example, previous research in the expectation literature has shown that expectation disconfirmation can not only induce heightened alertness (i.e., increased attention to unexpected vs. anticipated occurrences) but also a drive to make sense of the disparity and seek an explanation for the unexpected event (Roese & Sherman, 2007). Understanding a system's decision-making process is more cognitively effortful and may be particularly important when the system's performance is unreliable: Users are more likely to trust a system if they have a more accurate understanding of the process by which an AI system arrives at its decisions (Lee & Moray, 1992). Therefore, users may perceive a higher degree of understanding of an AI system if it provides more algorithmic transparency when their expectations are disconfirmed.

H4. When the AI system disconfirms users' expectations, greater transparency will enhance perceived understanding of the AI system, which will lead to a) greater cognitive trust, and (b) stronger behavioral intentions in using the AI system in the future.

2. Method

To examine the hypotheses and research questions, we conducted a 3 (Algorithmic transparency: high vs. low vs. absent) $\times$ 3 (Expectation confirmation: positive vs. simple vs. negative) between-subjects online experiment.

2.1. Participants

Upon receiving Institutional Review Board (IRB) approval, we recruited 284 participants from CloudResearch (Litman et al., 2017), which is an online crowdsourcing platform that helps to recruit high-quality participants from MTurk by screening out inattentive and fraudulent MTurk respondents (Hauser et al., 2023). We restricted our participants to those living in the U.S. who were currently not in a relationship, with over 500 approved Human Intelligence Tasks (HITs) on the platform and an Approval Rate above 95%. After excluding participants who did not complete the questionnaire ($n = 16$) and failed the attention check question ($n = 41$), our final sample comprised 227 participants.

The sample had an average age of 34.57 years ($SD = 7.22$, range: 19–55), with 58.1% female ($n = 132$) and 85.8% White/Caucasian ($n = 172$). 37.9% ($n = 86$) had attained a four-year bachelor's degree or higher. Participants generally reported high familiarity with existing online dating sites ($M = 5.31$, $SD = 1.65$) and moderately high interest in seeking relationships ($M = 4.08$, $SD = 2.15$), with no significant differences across conditions ($p_s > 0.14$). An *a priori* power analysis using G*Power (Faul et al., 2007) indicated a minimum sample size of 196 to achieve a power of 0.8 with a medium effect size of 0.25 (Cohen, 1977) and α of 0.05, which our final sample exceeded.

2.2. Stimuli

In this study, we created a fictitious online dating platform called *SmartMatch* and informed participants that it operates using algorithmic matching.

Online Dating Profiles. As perceived attractiveness of profile is one of the key attributes in online dating performance, we first conducted a pretest with 60 participants from CloudResearch (32 Male, 28 Female; $Mean_{age} = 32.07$). Following the procedures of Ristić et al. (2017), we curated two initial profiles (20 per gender) using photos from a free stock image database. Participants rated each profile's perceived attractiveness on a 7-point scale (1 = not at all attractive, 7 = extremely attractive). The ratings for both photo arrays, females and males, were normally distributed. We chose the mean attractiveness score and standard deviation (SD) to discern a neutral level of attractiveness ($Mean_{men} = 4.19$, $Mean_{women} = 5$). We then selected the pictures above the mean score (i.e., more attractive) and the ones below the mean score (i.e., less attractive) to use in the performance manipulation, respectively. When multiple photos met these criteria, we prioritized those with the smallest standard deviation to ensure rating consistency.

Algorithmic Transparency Manipulation. In the *low transparency* condition, the site only disclosed the algorithm's name and a single sentence indicating that it "analyzes and matches individuals according to scientific principles" without specifying the criteria (Fig. 1a). In the *high transparency* condition, the site disclosed the scientific principles behind the matching based on its algorithms. The algorithm was named "Duet Total Compatibility System," which analyzes and matches individuals according to the scientific principles of similarity and complementarity (see Sprecher, 2011). Participants were also informed about the criteria (namely, the six personality factors) that the algorithms adopted (Fig. 1b). In the *no transparency* condition, the site did not provide any information about how it recommends potential matches to users.

a. Low Transparency



About our algorithm

Based on how you answered a series of questions about yourself, we get to know you properly. And this is where the dimensions of compatibility come in. Based on your personality, our Duet Total Compatibility System of SmartMatch analyzes and matches individuals according to scientific principles. We generate Top Picks based on our algorithm findings.

b. High Transparency



About our algorithm

After years of studying the science of love, our academic relationship experts have carried out numerous studies of happy couples, and they have worked out this algorithm to predict how well two people are suited for each other.

Based on how you answered a series of questions about yourself, we get to know you properly. And this is where the dimensions of compatibility come in. Our **Duet Total Compatibility System** of SmartMatch analyzes and matches individuals according to the scientific principles of both similarity and complementarity.

There are 6 **different personality factors** that influence how well two people are suited for each other:

1. Romantic impulsivity
2. Personal energy
3. Predictability
4. Flexibility
5. Decision-making style
6. Emotionality

We recommend compatible matches based on our findings that couples are more compatible

- when they are similar on romantic impulsivity, personal energy and predictability,
- and when they are different on flexibility, decision-making style and emotionality.

We generate Top Picks based on our algorithm findings.

Fig. 1. Transparency manipulation.

Performance Expectations Manipulation. One indicator of an online dating site's performance is the number of attractive partners it can find (Courtois & Timmermans, 2018), and users hold expectations regarding the number of dates they will get to choose from on a dating site (Lenton et al., 2008). That is, the number of attractive dates recommended by the system may either confirm or violate users' expectations. To set participants' expectations for normal system performance, the site informed users that "Our system normally generates **five** 'Top Picks' for each user" (Fig. 2). We selected the profiles pretested above the mean score as the Top Picks. Subsequently, on the Discover Page, every participant received ten matching recommendations, with the simple confirmation condition showing five Top Picks and five profiles with lower attractiveness scores (Fig. 3a). In the negative confirmation condition, only two Top Picks and eight low attractive profiles were displayed (Fig. 3b). In the positive confirmation condition, eight Top Picks and two low compatible dates were shown to participants (Fig. 3c).

2.3. Procedure

Upon consenting to participate in the study on Qualtrics, participants were first asked about their attitudes toward AI-powered recommendation systems (i.e., AI belief). After that, they were instructed to visit an online dating site called "SmartMatch.com" and were randomly assigned to one of the nine conditions aligned with their sexual orientation. Following the standard practices of online dating services, the site asked participants several basic questions, such as age and body type, and a series of personality questions to rate from "1-Not me" to "5-Totally me" (e.g., "I enjoy vibrant social events with lots of people"). The site then informed participants that it would find them ten potential dates on their "Discover Page," and normally they would receive five "Top Picks." After viewing the result page (which varied in the number of Top Picks based on the condition assigned), participants read information about the site's matching algorithm. Following this, the site provided a survey code for participants to input into Qualtrics for continuing with the remainder of the study. The code was deemed necessary for ensuring that all participants completed their interaction with the website. Participants then completed the remaining questionnaire, which captured mediators, outcome variables, and demographic information. Upon completion, they were debriefed and compensated.

2.4. Measurements

All measures were self-reported items on 7-point Likert scales, ranging from "1 = strongly disagree" to "7 = strongly agree" unless otherwise mentioned. These items were all drawn from established instruments. When adapting items to fit the online-dating context, we made only minimal wording adjustments (e.g., replacing domain-specific terms with context-appropriate ones) while preserving the underlying construct definitions.

To check if the system performance expectation manipulation was

effective, we asked participants to rate the statement of "the online dating site performed 1 = worse than expected, 4 = the same as expected, 7 = better than expected," adapted from Bhattacharjee (2001). To check the manipulation of levels of transparency, we asked participants to rate the following statement, "The online dating site provides detailed information about its algorithm to users," on a 7-point Likert scale (Sun & Sundar, 2022).

Three items were adapted from Delibasic et al. (2013) and Pu et al. (2012) to capture *perceived understanding* (e.g., "I understood why certain potential dates were recommended to me"). *Cognitive trust* was measured by items (e.g., "The online dating site performs reliably") adapted from Madsen and Gregor (2000), and *behavioral trust* by items from Belanche et al. (2012), e.g., "I would like to use SmartMatch.com when I'm looking for potential partners". We measured *behavioral intentions toward the Top Picks* via three items adapted from Sharabi (2021) (e.g., "I want to talk to them soon").

Lastly, to rule out the confounding effects of predispositions toward algorithm-powered recommendation systems, we controlled for *AI belief* (e.g., "A mathematical formula can predict who I will be attracted to") adapted from Sharabi (2021). To assess whether participants perceived the online dating site as realistic, they were asked to rate the online dating site "SmartMatch" on three semantic-differential items (Bechwati & Xia, 2003): 'unrealistic–realistic', 'very unlikely in real life–very likely in real life', and 'not believable–very believable'. Table 1 reports the complete measurement information.

3. Results

In order to confirm measurement validity, we first assessed the measures with confirmatory factor analysis (CFA). The results showed a satisfactory fit of the model to the data: $\chi^2/df = 1.92$, $p < .001$, RMSEA = 0.07, 95 % CI [0.06, 0.07], CFI = 0.93 (Browne & Cudeck, 1993). All factor loadings were higher than 0.5 and significant (p 's < 0.01). The results also confirmed that composite reliabilities (CR) for each construct were higher than 0.85, exceeding the minimum criterion of 0.70, confirming convergent validity (Fornell & Larcker, 1981). All AVE (Average Variance Extracted) estimates were greater than the squared inter-construct correlations, confirming discriminant validity (Table 2).

3.1. Manipulation checks

A pair of 3 x 3 ANCOVAs, controlling for AI beliefs, confirmed that our manipulations were successful. The main effect of system performance expectation on the expectation manipulation-check item was significant: $F(2, 217) = 26.55$, $p < .001$, $\eta^2 = 0.20$. Participants in the positive confirmation condition ($M = 5.82$, $SE = 0.21$) rated the system's performance significantly better than those in the simple confirmation ($M = 4.96$, $SE = 0.22$) and negative confirmation ($M = 3.64$, $SE = 0.22$) conditions. The main effect of transparency disclosure on the transparency manipulation-check item was also significant, $F(2, 217) = 103.01$, $p < .001$, $\eta^2 = 0.54$. Participants in the high transparency condition ($M = 4.37$, $SE = 0.10$) rated the system as disclosing significantly more information about algorithms than those in low transparency ($M = 2.08$, $SE = 0.11$) and control conditions ($M = 1.29$, $SE = 0.11$).

3.2. Hypotheses testing and research questions

H1 predicted that when the site positively disconfirmed users' expectations, people would have higher cognitive trust and behavioral intentions than when the site simply confirmed or negatively disconfirmed their expectations. However, an ANCOVA analysis did not find any significant main effects for system performance expectation on cognitive trust, $F(2, 217) = 1.96$, $p = .143$, $\eta^2 = 0.02$, or behavioral intentions toward the site, $F(2, 217) = 1.02$, $p = .36$, $\eta^2 = 0.01$. Therefore, H1 did not receive support from our data.

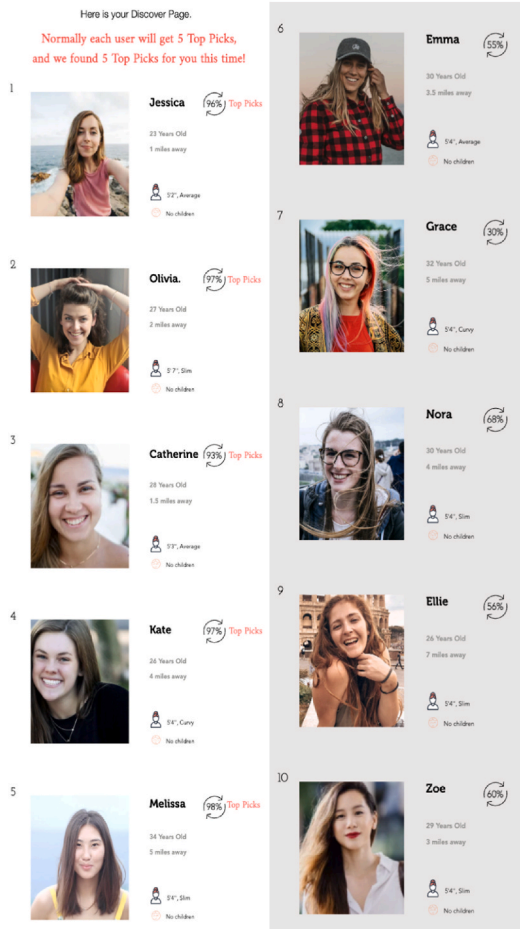
SmartMatch 

★ How many **Top Picks** to expect this time?

While the frequency may vary, our system normally generates **5 "Top Picks"** for each user.

Fig. 2. System performance expectation manipulation.

a. Simple Confirmation



b. Negative Disconfirmation

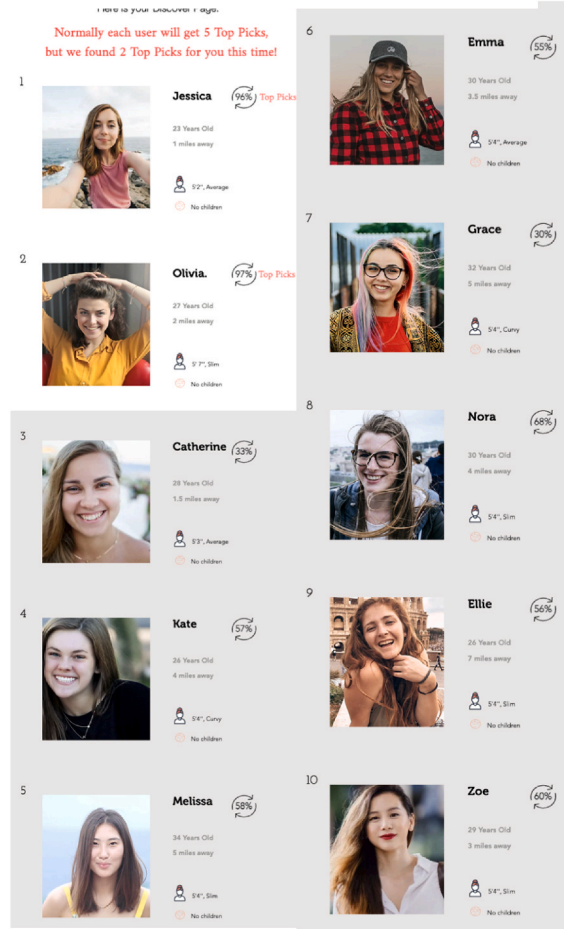


Fig. 3. Discover page (matching results with female candidates).

H2 hypothesized the positive effects of greater transparency on perceived understanding. An ANCOVA showed that the main effect of transparency on perceived understanding was significant, $F(2, 217) = 6.445, p = .002, \eta^2 = 0.06$. Specifically, a post-hoc analysis showed that participants who received more algorithmic transparency reported significantly higher perceived understanding ($M = 4.19, SE = 0.18$) compared to low algorithmic transparency ($M = 3.99, SE = 0.18$) and no transparency conditions ($M = 3.35, SE = 0.17$), supporting H2.

To examine the mediating role of perceived understanding proposed in H3, we conducted the mediation analysis using SPSS PROCESS macro (Model 4) with 5000 bootstrap samples (Hayes, 2017). The indirect effect was significant for no versus low transparency ($b = -0.27, 95\% \text{ CI} [-0.49, -0.08]$) but not for high versus low transparency ($b = 0.04, 95\% \text{ CI} [-0.18, 0.26]$). Direct effects on cognitive trust were non-significant for both comparisons ($ps > 0.138$), suggesting full mediation through perceived understanding. Overall, H3 was supported.

We conducted moderated mediation analyses using PROCESS Model 7 (Hayes, 2017) with 5000 bootstrap samples to examine whether AI transparency's effect on trust outcomes through perceived understanding varies by expectation disconfirmation (H4). The analysis included transparency levels as the independent variable ($X: 0 = \text{low}, 1 = \text{high}$), perceived understanding as the mediator, expectation disconfirmation as the moderator ($W: 0 = \text{disconfirmed}, 1 = \text{confirmed}$), and AI beliefs as a covariate. As predicted, the indices of moderated mediation were significant on cognitive trust ($b = -0.51, SE = 0.21, 95\% \text{ CI} [-0.92,$

$-0.10]$) and behavioral intentions toward the site ($b = -0.72, SE = 0.29, 95\% \text{ CI} [-1.30, -0.17]$). Specifically, when the system confirmed users' expectations by generating the normal number of Top Picks, disclosing less transparency to users led to a greater level of understandability ($M = 4.54$) compared to the high transparency condition (Fig. 4, middle; $M = 3.74, p < .001$). When the system violated user expectancy by either generating more Top Picks (i.e., positive confirmation; Fig. 4, right) or fewer Top Picks (i.e., negative confirmation; Fig. 4, left), the high-transparency disclosure condition led to greater perceived understanding (compared to the low-transparency condition (Table 3). Overall, H4 was supported.

4. Discussion

As algorithms significantly influence our social and romantic opportunities, we know little about how to effectively communicate the inner workings of these algorithms to users, particularly considering diverse user expectations. This study investigated how algorithmic explanations interact with user expectations to influence trust and satisfaction in AI-mediated dating platforms. By exploring these dynamics, it extends the ECM and HSM to human-AI interactions and examines how transparency enhances user trust when AI performance disconfirms user expectations. Practically, this research offers insights for a more ethical and user-centric design in AI-mediated social and romantic interactions.

c. Positive Disconfirmation

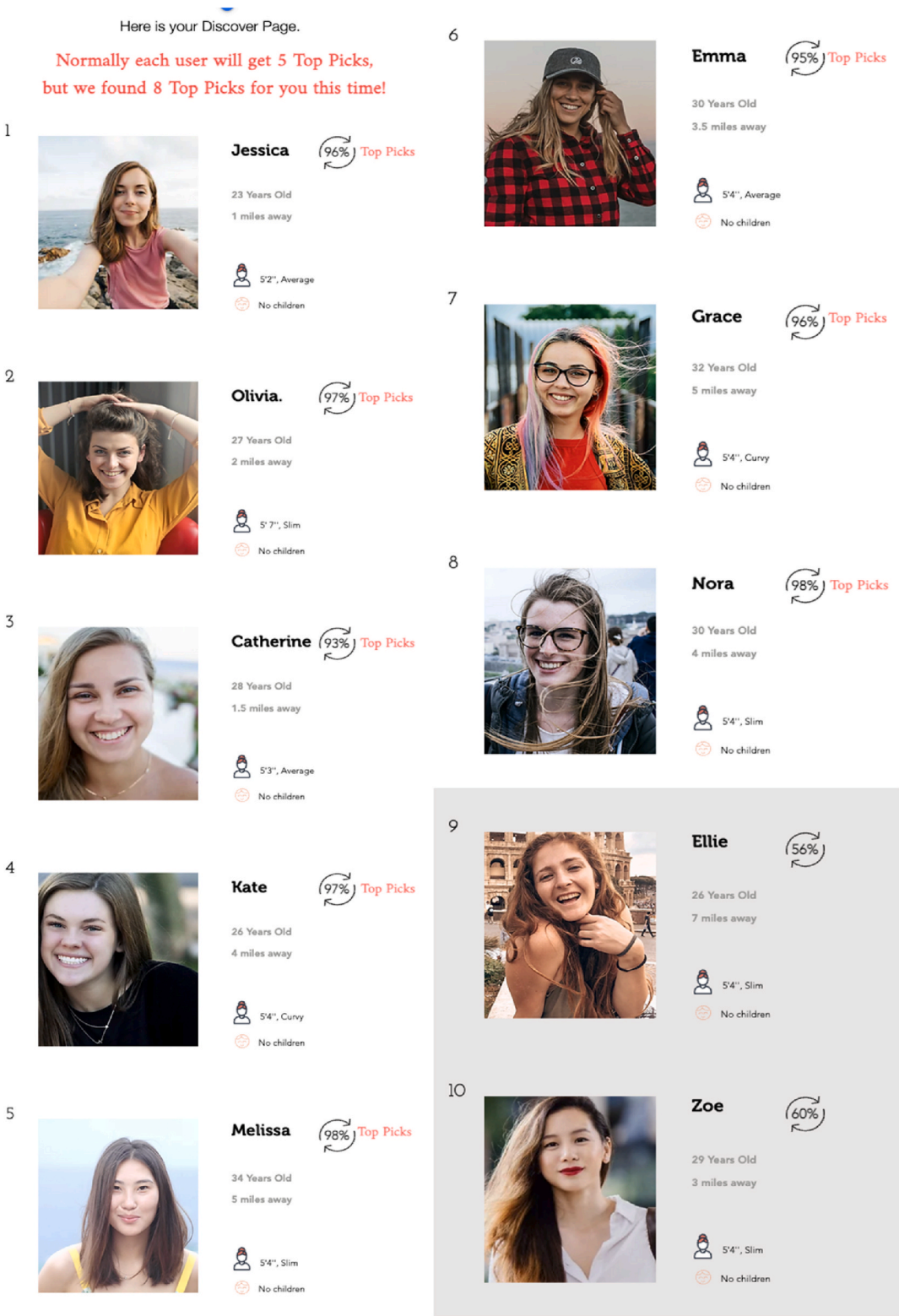


Fig. 3. (continued).

Table 1
Measurement scales.

Variables	Items	α	Mean (SD)
Perceived Understanding (Delibasic et al., 2013; Pu et al., 2012)	1. I understood why certain potential dates were recommended to me	0.93	3.83 (1.56)
	2. The system helps me understand why these online dating users were recommended to me		
	3. I understand how the online dating site assists me with the decisions I have to make		
Cognitive Trust (Madsen & Gregor, 2000)	1. The online dating site performs reliably	0.93	4.49 (1.21)
	2. I can rely on the online dating site to function properly		
	3. The online dating site uses appropriate methods to reach decisions		
	4. The online dating site has sound knowledge about how to recommend compatible dates		
	5. The online dating site correctly uses the information I enter		
Behavioral Intentions toward the Online Dating Site (Belanche et al., 2012)	I would like to:	0.95	3.87 (1.66)
	1. Spend more time on the site SmartMatch.com		
	2. I would like to use SmartMatch.com when I'm looking for potential partners		
	3. I would recommend the site SmartMatch.com to my friends		
AI Belief (Covariate; Sharabi, 2021)	1. The matching algorithms really work	0.92	3.88 (1.25)
	2. I would trust a matching algorithm to find me a partner		
	3. Matching algorithms lead to more successful relationships		
	4. A mathematical formula can predict who I will be attracted to		
	5. Matching algorithms are better at finding me a partner than I am		
	6. Matching algorithms provide me with better quality partners		
	7. Matching algorithms are more effective than traditional ways of meeting people		
Scenario Realism (Bechwati & Xia, 2003)	The online dating site of "SmartMatch" was	0.91	4.98 (1.34)
	1. Unrealistic – Realistic		
	2. Very unlikely in real life – Very likely in real life		
	3. Not believable – Very believable		

Table 2
Measurement properties and construct correlations.

Construct	Items	CR	AVE	α	1	2	3
1. Perceived Understanding	3	0.91	0.70	0.90	0.84		
2. Cognitive Trust	5	0.92	0.68	0.91	0.59	0.82	
3. Behavioral Intentions (Site)	3	0.89	0.65	0.88	0.54	0.65	0.81

4.1. Theoretical implications

First, we extend ECM in AI contexts by demonstrating that expectation confirmation alone may not drive trust development. Contrary to traditional ECM predictions, our results showed that positive disconfirmation in AI performance expectation did not automatically generate greater trust, challenging the previous work of ECM in HCI literature (e.g., Bhattacharjee et al., 2008; Burgoon et al., 2016). The null findings may be interpreted in light of the significant interaction effects of AI transparency and performance expectations. In fact, when AI systems outperform expectations, users may experience increased curiosity about the system's internal mechanisms. This curiosity, in turn, creates a need for transparency to explain such inconsistencies, as evidenced by our significant moderated mediation effect via perceived understanding of the AI system.

The study also contributes to the current literature of explainable AI (XAI) (Gunning et al., 2019) by shedding light on the psychological mechanism through which transparency affects trust. Previous studies have demonstrated that users showed higher trust in the AI system with the mere presence of transparency information compared to no transparency (Sundar et al., 2020). The results of our study extend previous literature by establishing perceived understanding as a full mediator that explains the positive effects of providing detailed algorithmic transparency. The finding bridges theoretical gaps between transparency design and trust outcomes, suggesting that effective transparency must prioritize user understanding rather than merely providing information.

Most significantly, our results reveal a nuanced interaction between transparency and expectation confirmation that advances both theoretical domains. The finding that the effectiveness of transparency can be more salient based on expectation confirmation suggests a dynamic rather than static role of transparency in trust formation. When AI systems behave as expected, minimal transparency may suffice for trust development. This suggests that users are more likely to engage in heuristic processing when the system meets their expectations, preferring simpler explanations that will not overwhelm them or interfere with their user experience (Bunt et al., 2012) without providing substantive value (Bunt et al., 2012), or leading to more questions (Kizilcec, 2016). As Banas et al. (2022) showed, providing transparency might not always be effective if users gain little new information from such efforts. In contrast, when expectations are disconfirmed (either positively or negatively), more detailed transparency could increase users' understanding and trust in the system, which could be due to a shift towards more systematic processing of the AI system's operations. Our study further adds to the literature about communicating algorithmic transparency to lay users by identifying perceived understanding as the key mechanism underlying the combined effect of AI performance expectations and transparency on trust outcomes. This theoretical extension calls for a reevaluation of Explainable AI (XAI) strategies, emphasizing the need to optimize explanations for user comprehension and to consider the context of user expectations when designing AI transparency measures.

The key mechanism underlying the combined effect of AI performance and transparency appears to be perceived understandability of the AI system. This finding extends the existing trust literature by revealing a new antecedent of trust. Previous studies have mainly examined perceptions of trustworthiness, benevolence, and integrity as antecedents for developing trust (Hwang & Kim, 2007; Jarvenpaa et al., 1998; Mayer et al., 1995). Our study suggests that users' perceptions of understandability constitute another crucial underlying factor in the AI trust-building process. Perceived understandability was operationalized as perceived comprehension of how the AI-driven site reaches its recommendations. It is affected by system performance and the information disclosed by the system, as supported by the significant interaction effect of the two IVs. This shows that transparency information alone does not translate to understandability, which has been identified as one of the

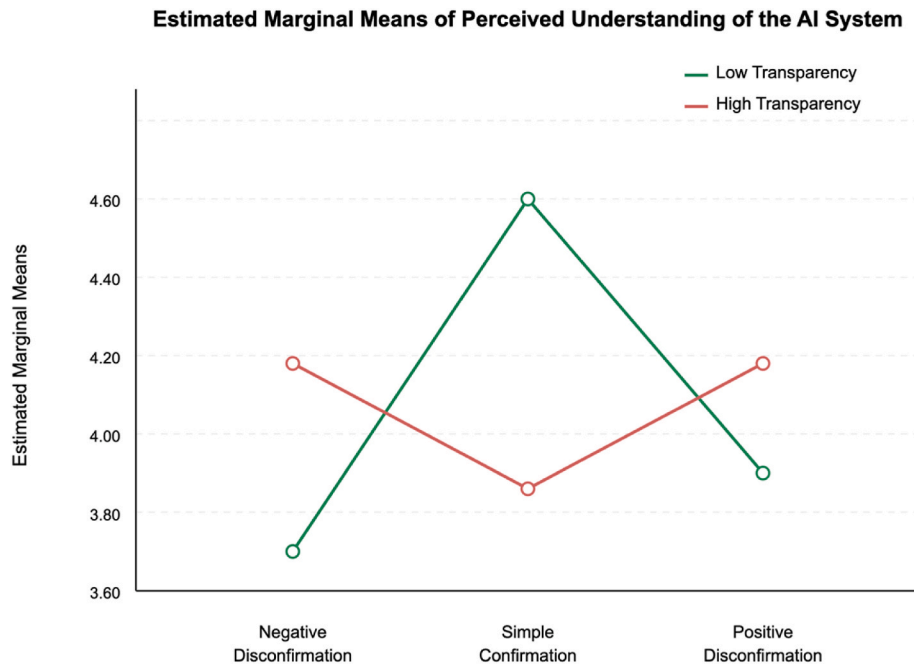


Fig. 4. Interaction effects between performance expectations and transparency level on perceived understanding of the AI system.

Table 3

Means and SE for the two-way Interaction of Performance Expectation and Transparency on Perceived Understanding.

Performance Expectation	AI Transparency	<i>M</i>	<i>SE</i>
Simple Confirmation	Low Transparency	4.54	0.30
	High Transparency	3.74	0.30
Negative Confirmation	Low Transparency	3.48	0.35
	High Transparency	4.35	0.38
Positive Confirmation	Low Transparency	4.00	0.35
	High Transparency	4.51	0.31

Note: Means with no subscript in common differ at $p < .05$ using Bonferroni post hoc comparisons.

major challenges in XAI (Gunning et al., 2019). As users look for resources to understand automated systems, when discrepancies occur (Muir & Moray, 1996), systems' performance levels may moderate the effects of transparency on understandability. Our study provides insight into the current discourse about XAI, suggesting that different transparency strategies are needed, taking users' expectations into account. Specifically, more transparency is needed when users' expectations of system performance are violated, while such information may backfire when their expectations of AI systems are met.

4.2. Practical implications

Our findings offer actionable insights for AI system design and deployment. Designers should implement adaptive transparency mechanisms based on expectation disconfirmation. When systems perform unexpectedly, more detailed explanations become crucial. This could be operationalized through user feedback mechanisms, such as having users rate recommendations as "better than," "similar to," or "worse than" expected, with transparency levels adjusting accordingly.

For UX designers, our suggestion is to prioritize understandability over information volume. The critical role of perceived understanding suggests implementing features that enhance comprehension. After providing algorithmic transparency, systems could incorporate simple checkpoints like "How well do you understand how the recommendations were made?" Such questions can increase algorithm awareness and encourage engagement with system explanations.

4.3. Limitations and future directions

This study, while offering valuable insights, has several methodological limitations that suggest promising directions for future research. First, our participants demonstrated high familiarity with online dating platforms ($M = 5.31$, $SD = 1.65$), potentially limiting generalizability to first-time users or individuals less experienced with algorithmic recommendation systems. The sample was also predominantly White/Caucasian (85.8 %) and recruited through online platforms, which may not represent diverse populations or individuals with different levels of digital literacy. Future research should examine these dynamics across more heterogeneous samples, particularly focusing on populations who may be less familiar with AI-driven platforms or who may hold different cultural expectations about algorithmic decision-making. Second, our expectation manipulation focused exclusively on quantitative performance differences (number of "Top Picks" provided). While effective for our research purposes, this approach does not capture other dimensions along which AI systems might confirm or violate user expectations, such as explanation quality, system responsiveness, or the depth of recommendation relevance. Lastly, our study examined a single interaction episode, whereas expectation confirmation is likely a dynamic process that evolves through repeated interactions. Longitudinal research is necessary for investigating how trust develops as users gain more experience with AI systems.

5. Conclusion

As AI continues to play an increasingly significant role in mediating our social interactions and relationships, understanding how to foster appropriate trust through effective transparency becomes crucial. Extending ECM to the human-AI interaction context, this study advances our understanding of algorithmic transparency and trust in AI systems by examining the moderating role of user expectation. Our findings revealed that while expectation confirmation alone did not affect trust in the AI system, communicating detailed algorithmic transparency when user expectation was disconfirmed significantly enhanced perceived understanding of the system and subsequent trust. These results highlight the dynamic relationship between user expectations and algorithmic transparency, suggesting that effective explanations must

prioritize comprehension over mere information disclosure. These insights hold important implications for the human-AI literature and provide practical implications for designing explainable and trustworthy AI systems.

CRedit authorship contribution statement

Yuan Sun: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mengqi (Maggie) Liao:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Conceptualization. **S. Shyam Sundar:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization, Methodology, Validation. **Joseph B. Walther:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization, Methodology, Validation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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